

Daily Task Completion Report

Admin-Side Refund Processing and Notifications

Date: 03/09/2026

Task: Admin Return Lifecycle, Refund Processing, Inventory Management & Notifications

Status: Completed

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Overview

Today, I completed the **admin-side return and refund workflow** for the e-commerce application. The implementation covers the complete return lifecycle from admin review to approval/rejection, inventory updates, payment refunds, and user notifications.

1. Admin Return APIs

Implemented admin APIs to manage customer return requests:

- **GET /admin/returns** – Retrieves return requests for admin review.
- **POST /admin/returns/{id}/approve** – Allows the admin to approve a return request.
- **POST /admin/returns/{id}/reject** – Allows the admin to reject a return request.

These APIs provide the admin with control over the complete return-processing workflow.

2. Return Status Handling

Implemented proper status transitions based on the admin's action:

- **Approved → Returned**
- **Rejected → Rejected**
- **Refund completed → Refunded**

Admin

GET	/admin/users	Get All Users	🔒	▼
GET	/admin/analytics	Get Admin Analytics	🔒	▼
GET	/admin/returns	Get All Returns	🔒	▼
POST	/admin/returns/{return_id}/approve	Approve Return	🔒	▼
POST	/admin/returns/{return_id}/reject	Reject Return	🔒	▼
POST	/admin/returns/{return_id}/refund	Refund Return	🔒	▼

Admin dashboard

The screenshot displays a REST client interface on the left and a WebSocket console on the right. The REST client shows a PUT request to `http://127.0.0.1:8000/orders/65/status?status=Delivered` with a 200 status code. The response body is a JSON object containing order details. The WebSocket console on the right shows a series of messages, including order status updates and notifications, indicating that the request was successfully processed and the system is sending real-time updates.

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/orders/65/status?status=Delivered' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/orders/65/status?status=Delivered

Server response

Code	Details
200	Response body <pre>{ "message": "Order status updated successfully", "order": { "id": 65, "user_id": 5, "total_amount": 128998, "payment_status": "Paid", "order_status": "Delivered", "status": "Delivered", "created_at": "2026-08-24T17:39:37" }, "email_sent": true, "websocket_sent": true, "notification_websocket_sent": true }</pre> Response headers

WebSocket console messages:

- WebSocket message: VM163:10
{event: 'order_status_updated', order_id: 65, status: 'Delivered', message: 'Your order #65 is now Delivered.'}
- Order updated: VM163:13
{event: 'order_status_updated', order_id: 65, status: 'Delivered', message: 'Your order #65 is now Delivered.'}
- WebSocket message: VM163:10
{event: 'notification', notification_id: 49, type: 'order', message: 'Your order #65 is now Delivered.'}
- Notification: VM163:17
{event: 'notification', notification_id: 49, type: 'order', message: 'Your order #65 is now Delivered.'}
- WebSocket message: VM163:10
{event: 'order_status_updated', order_id: 65, status: 'Return Requested', message: 'Your order #65 is now Return Requested.'}
- Order updated: VM163:13
{event: 'order_status_updated', order_id: 65, status: 'Return Requested', message: 'Your order #65 is now Return Requested.'}
- WebSocket message: VM163:10
{event: 'notification', notification_id: 50, type: 'return', message: 'Your return request for order #65 has been submitted successfully.'}
- Notification: VM163:17
{event: 'notification', notification_id: 50, type: 'return', message: 'Your return request for order #65 has been submitted successfully.'}

Request delivered

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/orders/returns/3/approve' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/orders/returns/3/approve

Server response

Code Details

200

Response body

```
{
  "message": "Return approved",
  "return_request_id": 3,
  "status": "Approved",
  "order_status": "Returned"
}
```

Response headers

```
access-control-allow-credentials: true
content-length: 97
content-type: application/json
date: Mon, 31 Aug 2026 07:02:19 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

Refund Request Approved

Dimensions: Responsive 760 x 500 100% Fast 4G Save-Data default

Code Details

200

Response body

```
{
  "message": "Return approved successfully",
  "return_request_id": 5,
  "return_status": "approved",
  "order_id": 88,
  "order_status": "Returned",
  "inventory_restored": true,
  "restored_items": [
    {
      "product_id": 8,
      "product_name": "Laptop",
      "quantity_added": 2,
      "old_stock": 2,
      "new_stock": 4
    },
    {
      "product_id": 11,
      "product_name": "iPhone 15",
      "quantity_added": 3,
      "old_stock": 10,
      "new_stock": 13
    },
    {
      "product_id": 14,
      "product_name": "Laptop",
      "quantity_added": 3,
      "old_stock": 0,
      "new_stock": 3
    }
  ]
}
```

Response headers

Download

Elements Console Sources

4 hidden

- POST http://127.0.0.1:8000/admin/returns/88/approve 404 (Not Found)
- POST http://127.0.0.1:8000/admin/returns/84/approve 404 (Not Found)
- POST http://127.0.0.1:8000/admin/returns/7/approve 400 (Bad Request)

WebSocket message: VM519:10

{event: 'return_approved', return_id: 5, order_id: 88, status: 'approved', message: 'Your return request for order #88 has been approved.'}

WebSocket message: VM519:10

{event: 'order_status_updated', order_id: 88, status: 'Returned', message: 'Your order #88 is now Returned.'}

Order updated: VM519:13

{event: 'order_status_updated', order_id: 88, status: 'Returned', message: 'Your order #88 is now Returned.'}

WebSocket message: VM519:10

{event: 'notification', notification_id: 75, type: 'return', message: 'Your return request for Order #88 has been approved.'}

Notification: VM519:17

{event: 'notification', notification_id: 75, type: 'return', message: 'Your return request for Order #88 has been approved.'}

WebSocket message: VM519:10

{event: 'inventory_updated', product_id: 8, stock: 4, quantity_added: 2, message: 'Inventory updated for product #8.'}

WebSocket message: VM519:10

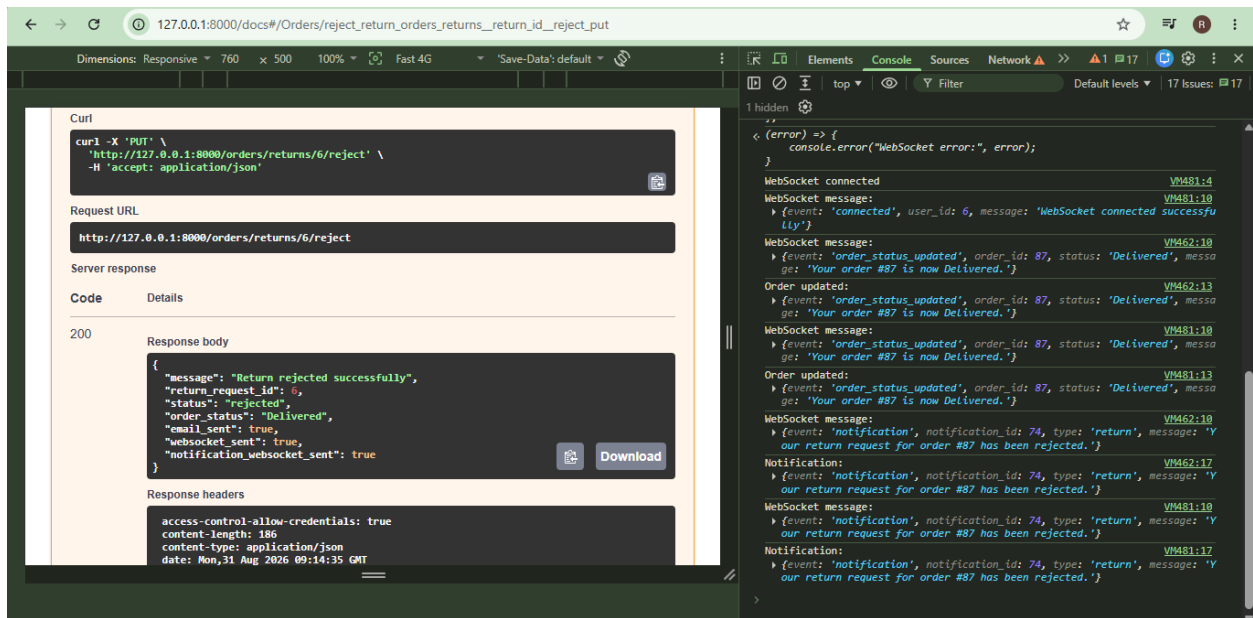
{event: 'inventory_updated', product_id: 11, stock: 13, quantity_added: 3, message: 'Inventory updated for product #11.'}

WebSocket message: VM519:10

{event: 'inventory_updated', product_id: 14, stock: 3, quantity_added: 3, message: 'Inventory updated for product #14.'}

WebSocket disconnected VM519:22

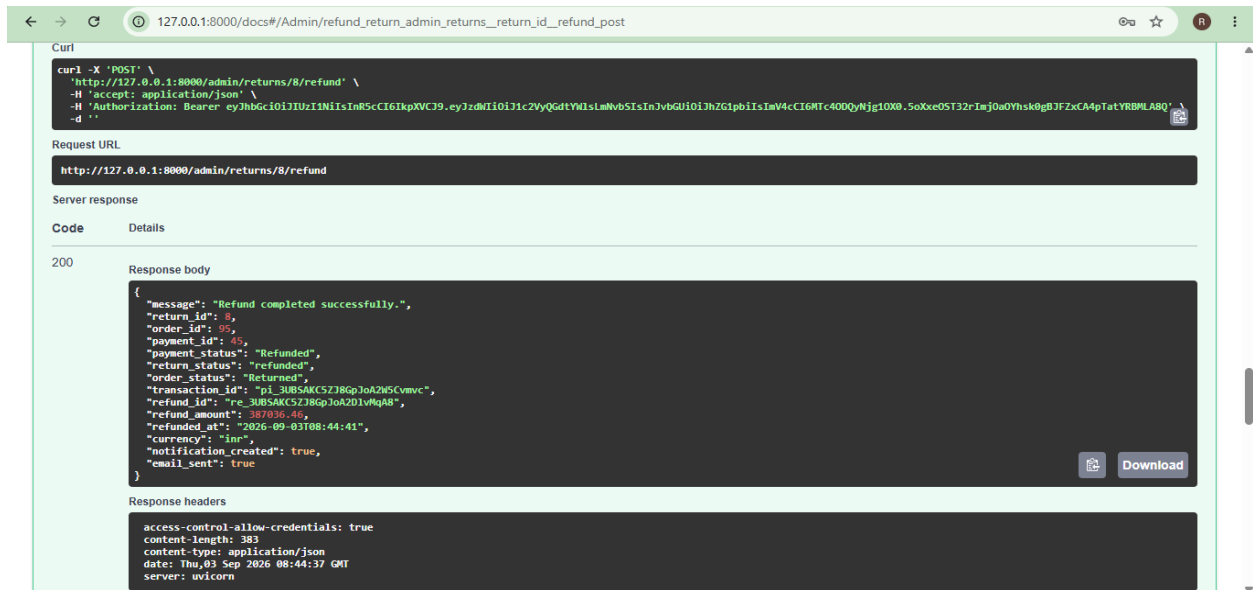
Quantity Increase after approve the request



Refund Request Rejected

3. Payment Refund

Integrated the **Stripe refund workflow** for processing refunds after an approved return. The payment status is updated after the refund process so that the system maintains accurate payment and refund information

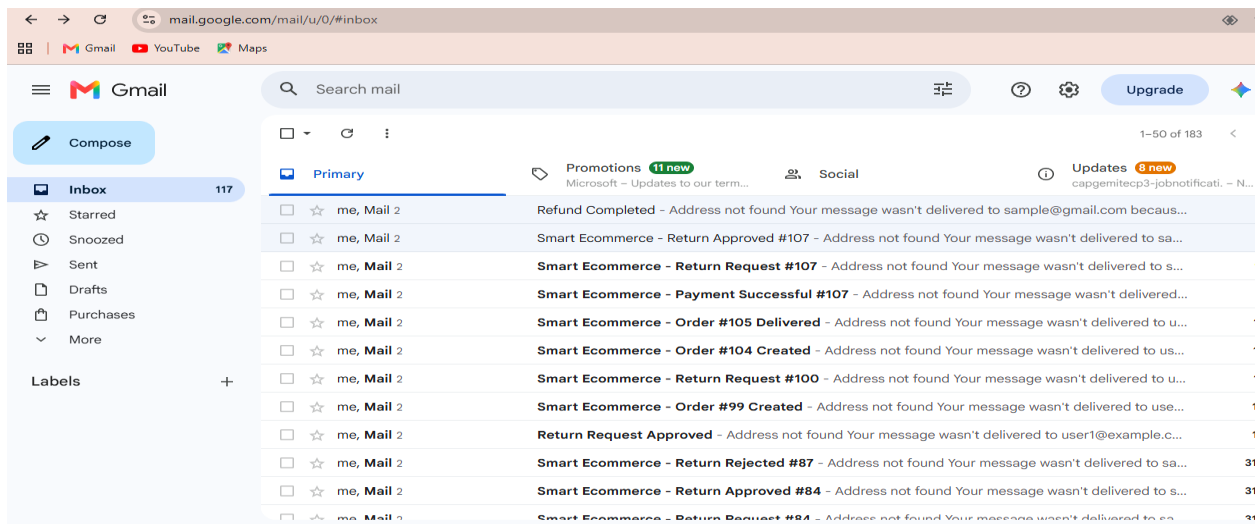


Refunded successfully

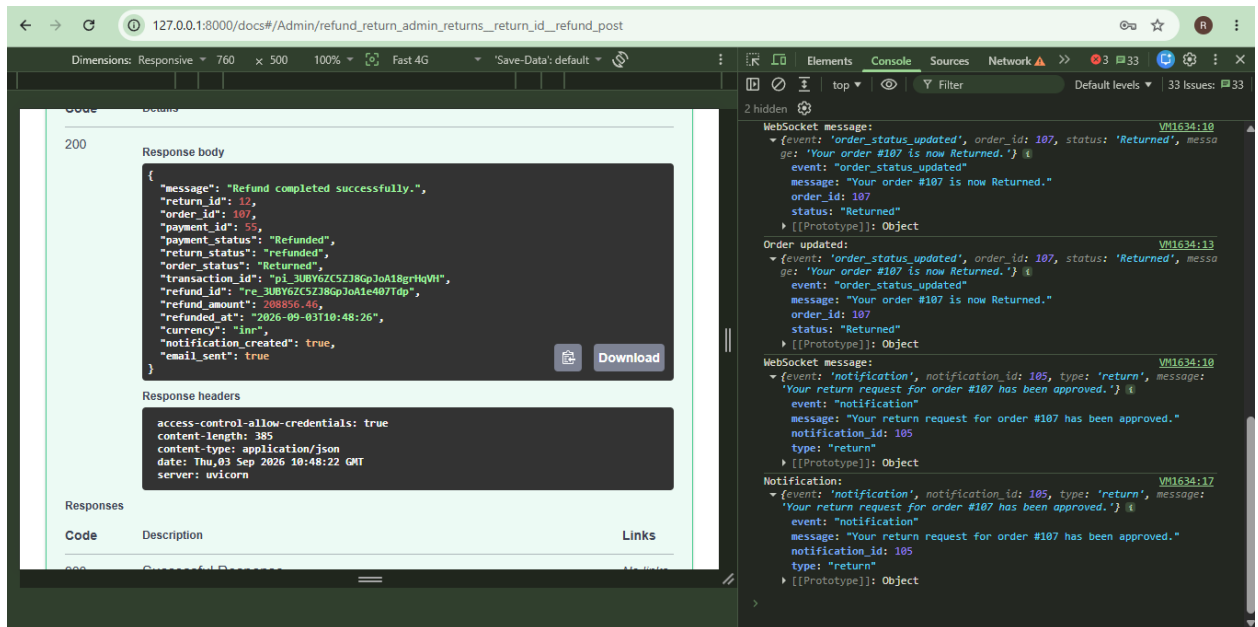
5. Notification System

Implemented notifications for important return and refund events. The system supports both **email notifications** and **in-app notifications**.

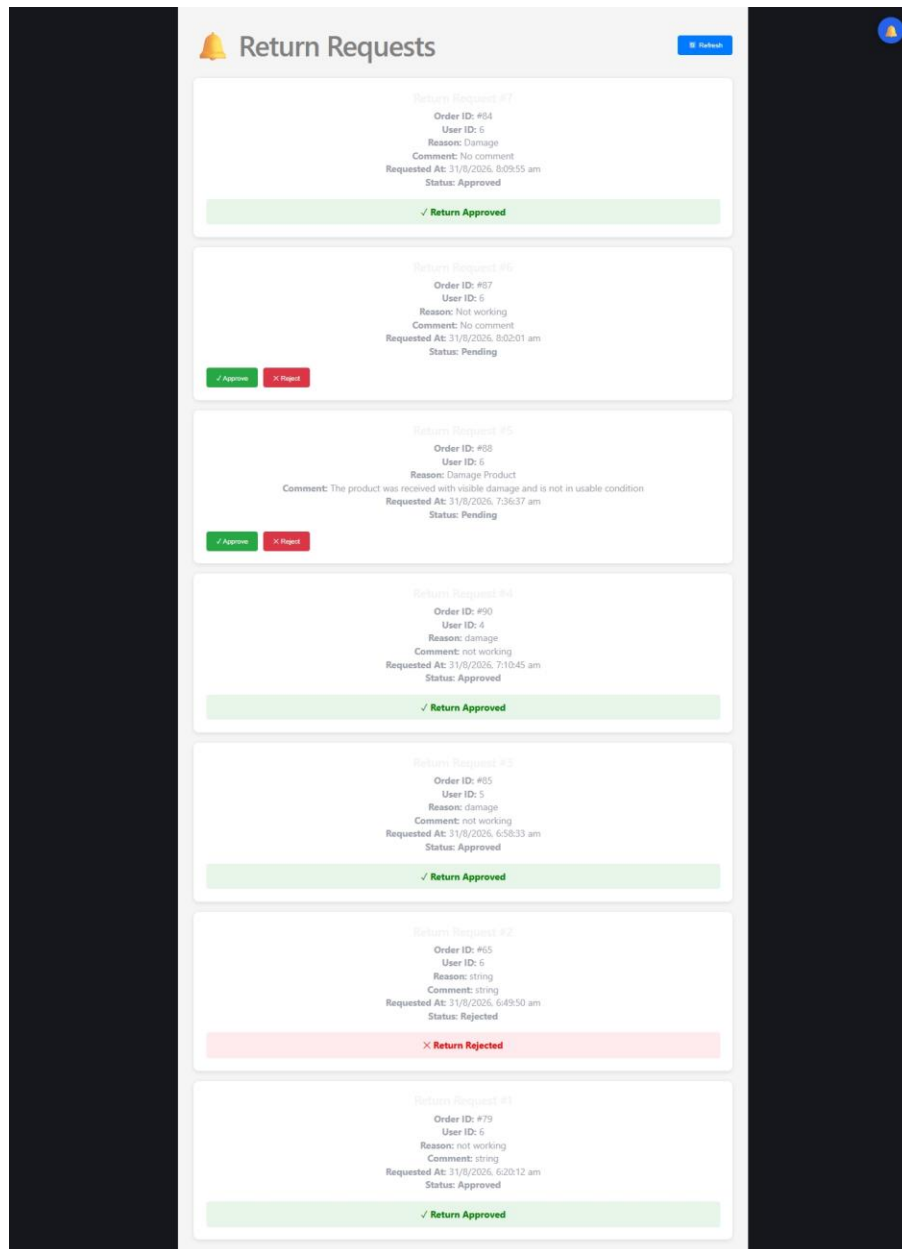
- Return approved
- Return rejected
- Refund completed



Email Notification



Websocketv Notification



Conclusion

The complete **admin return lifecycle and refund workflow** has been implemented successfully. Admins can review and process returns, inventory is updated after approval, Stripe refunds are processed, payment status is maintained, and customers receive notifications for important events. This provides a complete and structured return-to-refund process for the e-commerce platform.

Overall Status:  **Completed**